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Tarun@Tarun-PC ~
$ cd /cygdrive/c/Users/Tarun/Desktop/Docs/WSN_SampleProject/
Tarun@Tarun-PC /cygdrive/c/Users/Tarun/Desktop/Docs/WSN_SampleProject
$ gcc Main.c -o Main.exe
Tarun@Tarun-PC /cygdrive/c/Users/Tarun/Desktop/Docs/WSN_SampleProject
$ ./Main.exe
Running NetSim for Nodes      1
Running NetSim for Nodes      2
Tarun@Tarun-PC /cygdrive/c/Users/Tarun/Desktop/Docs/WSN_SampleProject
$

```

Fig. 2: Cygwin/Mingw output

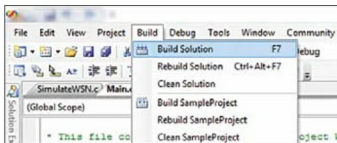


Fig. 3: Building solution in Visual Studio

in a backoff slot.

Discard probability. Probability of packet being discarded:

$P_{\text{discard}} = \text{Frames discarded} / \text{Frames transmitted (ignoring retransmission)}$

Payload and plots. Payload at application layer is 4 bytes. Overheads are set according to the standard (20IP+13MAC+6PHY). Using the Metrics.txt files obtained from the simulations, graphs are plotted. Below is an example:

Simulations are done in different scenarios for various nodes. Using Python's matplotlib module, you can plot a throughput plot. (To understand the significance of various plots, you may refer to WSN_White Paper.pdf file included in the DVD accompanying EFY Plus, and also at efymag.com).

From the plots, you can see that change in the backoff parameters (backoff multiplier and backoff exponent limits, i.e., macMinBE and MaxBE) leads to a higher throughput than with the 802.15.4 standard. As the attempt rate (β) increases, probability of a packet collision increases and so throughput decreases. Change in the backoff parameters decreases the attempt rate (β), as we notice in the plots, and therefore the collisions decrease leading to a higher throughput. This

decrease in collisions is evident from the discard probability graphs.

These simulation results validate the theoretical analysis of improved throughput (see

Fig. 1) with modifications to the 802.15.4 standard's back-off parameters.

Software tools

Following software tools have been used to develop and test this program:

1. 'C' compiler. The project has been tested with Visual Studio 2005, Cygwin and Mingw.
2. Python 2.7 (<http://www.python.org/download/>)
3. Matplotlib 1.1.1 compatible with Python 2.7 (<http://sourceforge.net/projects/matplotlib/files/matplotlib/matplotlib-1.1.1/>)
4. Numpy 1.6.2 compatible with Python 2.7 (<http://sourceforge.net/projects/numpy/files/>)

Project folders (INSN_SampleProject). The project contains following folders:

Configuration_Files_Division/. 'Configuration.xml' has been divided into several parts so that the user can create a custom XML file from 'C' code (Main.c) using the following parts which don't vary.

Dll/. All the dlls needed for running simulation are kept in this directory (taken from NetSim 6.1).

Documents/. This directory has all the documents corresponding to the

project, and also some good reference documents to understand IEEE 802.15.4 standard.

Main.c. This 'C' code loads the dlls in Dll directory. It also loads the functions (Sample_WSN_Project_NetSim) required for simulation, and declares the function (scenario) for creating 'Configuration.xml' files into 'NetSim_Temp_Files' directory.

PlotAll.py. For analysing the results after simulation, Python script has been used to plot the graphs. Numpy, fundamental package for scientific computing with Python, and Matplotlib, plotting library for Python, are used for plots.

NetSim_Temp_Files/. This directory is used as a temporary directory for storing the configuration, trace and metrics files obtained in simulation.

Debug/. This directory is used by Visual Studio to create the exe file.

Output/. After each simulation, the trace files, metrics files and other files that are needed for analysis are copied to this folder automatically by the program.

(The various files contained in each of these folders have been given in the Readme.pdf file in 'Document' folder.)

Testing procedure

CYGIN/MINGW. The procedure for building in 'Cygwin' is as follows:

1. Open 'SampleProject/Main.c' modify the code as per your requirement and save it.
2. Open 'Cygwin/MinGW' terminal. Now open 'SampleProject' directory and run the following commands to compile and run the simulation:
 - (i) gcc Main.c -o Main.exe
 - (ii) ./Main.exe

You can refer to Fig. 2 for 'Cygwin/MinGW' output.

Visual Studio. For building with Visual Studio, proceed as follows:

1. Open 'SampleProject.sln' located in 'SampleProject' folder with Visual Studio 2005 or higher version.
2. Modify the parameters as needed in 'SampleProject/Main.c' and save it.
3. Build the project using the menu 'Build→Build Solution' or use the shortcut F7 (see Fig. 3).